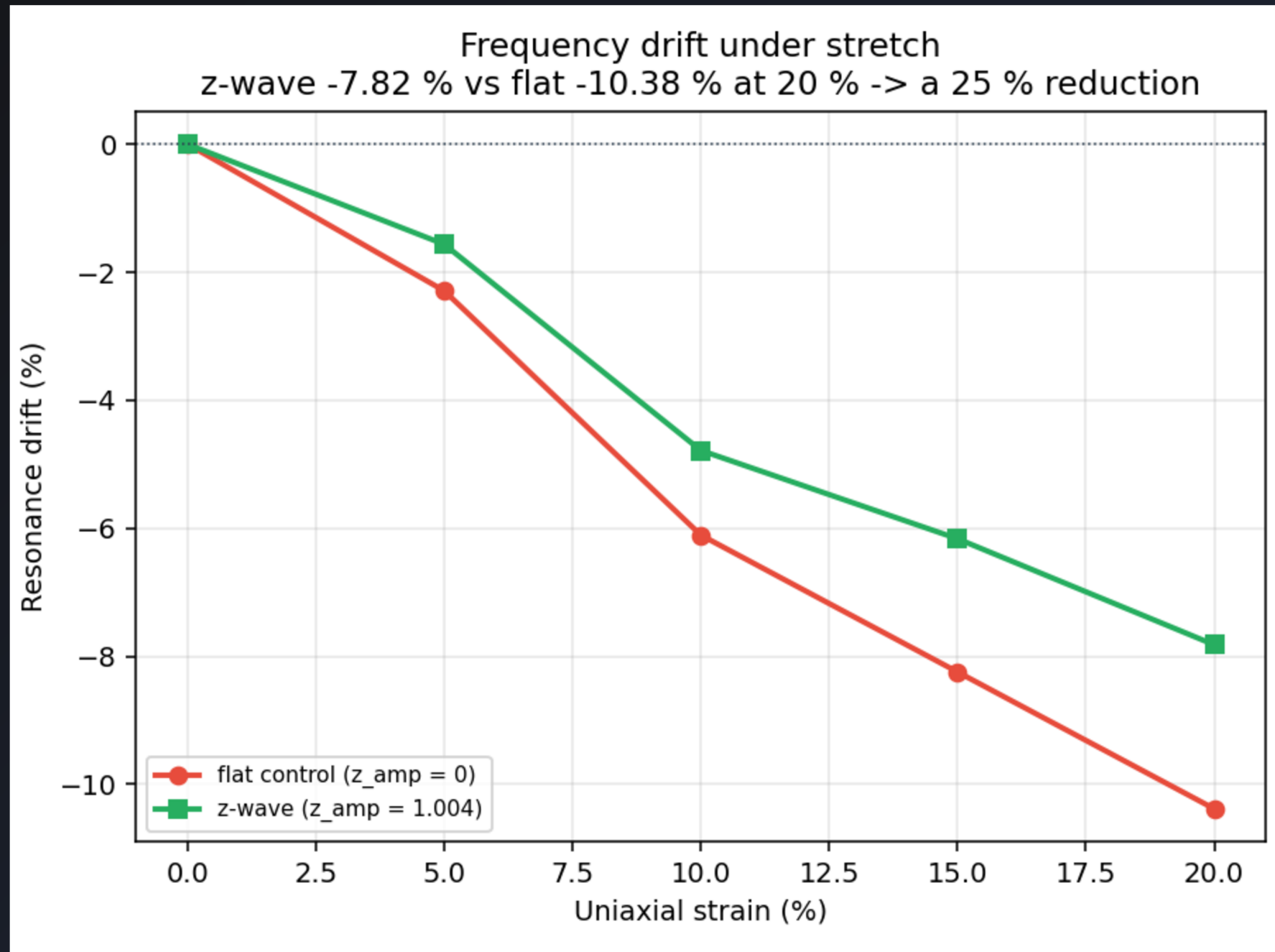


WORKING PRINCIPLES

Why a stretched antenna goes off band



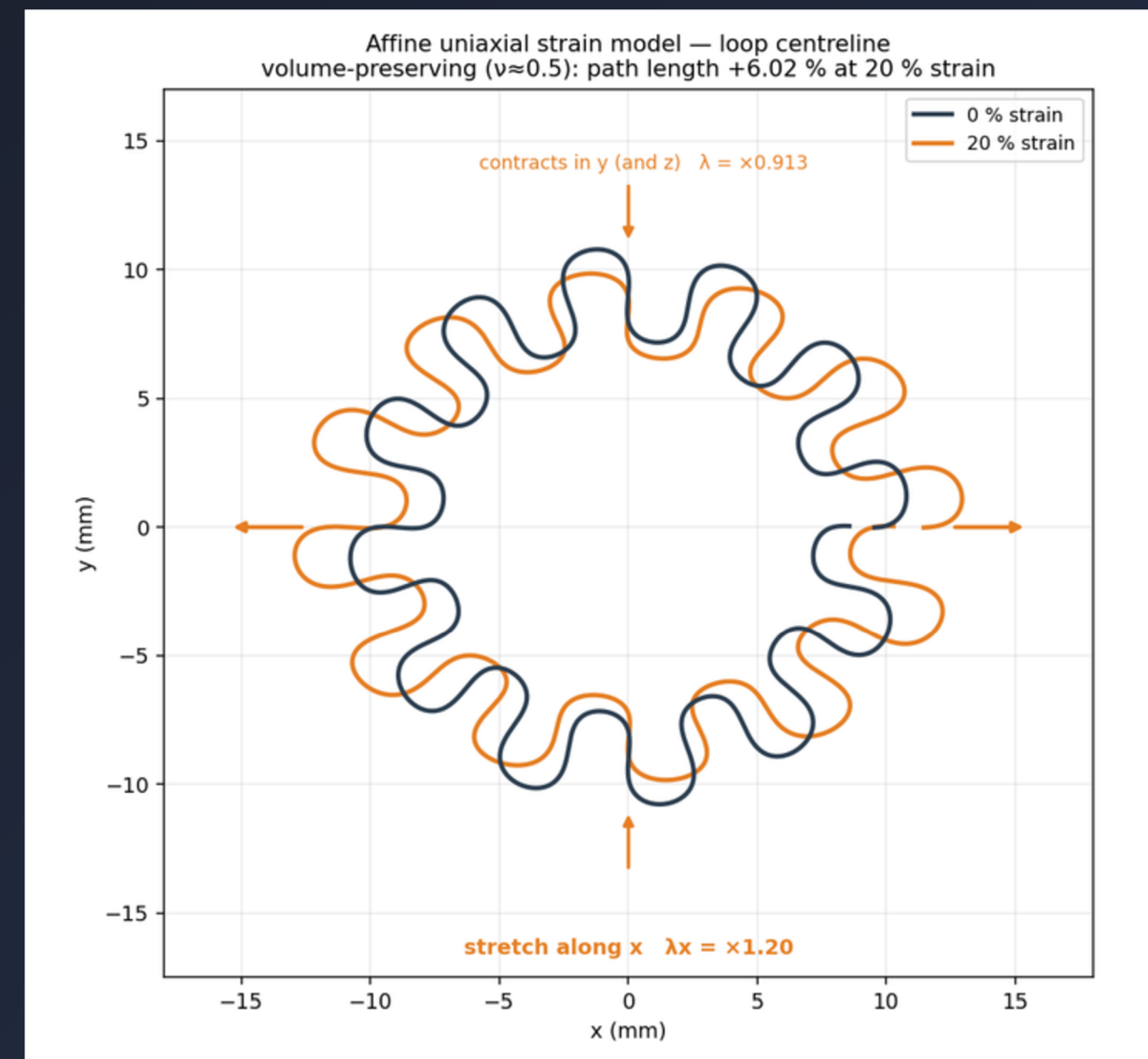
- A loop resonates when its perimeter $\approx$ one wavelength $\rightarrow f \propto 1/L$
- Stretch the tape, the liquid metal stretches with it, L grows, f falls
- At 20 % strain the in-plane path grows +6 % $\rightarrow$ -5.7 % in frequency
- BLE only gives ± 1.7 % of band $\rightarrow$ flat loop is off-band three times over

WORKING PRINCIPLES

How it is solved:

- Ecoflex is incompressible ($\nu \approx 0.5$). It is comfortable for human skin and when stretched sideways it gets thinner.
- So we run the channel up and down in z. Stretching flattens the wave, giving back length at almost exactly the rate the stretch takes it.

Result: -7.82 % vs -10.38 % for a flat control on the same substrate = 25 % less drift. So the antenna is strain-tolerant



top view of stretch